Evaluation Metrics

Measuring what actually matters

The problem

- 99% accuracy can be useless (1% fraud rate)
- Errors have different costs
- Need metrics that match the business goal

Confusion matrix

- TP: caught positives FN: missed positives
- FP: false alarms TN: correct negatives
- Layout here: [[TN, FP], [FN, TP]]

Precision, recall, F1

$$P = \frac{TP}{TP + FP} \quad R = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2PR}{P + R}$$

- Precision: how many alarms were real
- Recall: how many positives we found

Precision / recall / F1

```
def confusion_matrix(y_true, y_pred):
    """Binary confusion matrix [[TN, FP], [FN, TP]]."""
    y_true, y_pred = np.asarray(y_true), np.asarray(y_pred)
    tp = int(np.sum((y_true == 1) & (y_pred == 1)))
    tn = int(np.sum((y_true == 0) & (y_pred == 0)))
    fp = int(np.sum((y_true == 0) & (y_pred == 1)))
    fn = int(np.sum((y_true == 1) & (y_pred == 0)))
    return np.array([[tn, fp], [fn, tp]])

def precision_recall_f1(y_true, y_pred):
    (tn, fp), (fn, tp) = confusion_matrix(y_true, y_pred)
    p = tp / (tp + fp) if tp + fp else 0.0
    r = tp / (tp + fn) if tp + fn else 0.0
    f1 = 2 * p * r / (p + r) if p + r else 0.0
    return p, r, f1
```

ROC AUC: ranking quality

- AUC = P(random positive scores > random negative)
- Ties count half; threshold-free
- 0.5 = random, 1.0 = perfect ranking

AUC by pairwise comparison

```
def roc_auc(y_true, scores):
    """AUC = probability a random positive scores above a random negative (ties count half)."""
    y_true, scores = np.asarray(y_true), np.asarray(scores, dtype=float)
    pos, neg = scores[y_true == 1], scores[y_true == 0]
    if len(pos) == 0 or len(neg) == 0:
        raise ValueError("need both classes")
    diff = pos[:, None] - neg[None, :]
    return float(np.mean((diff > 0) + 0.5 * (diff == 0)))
```

Probabilistic and regression metrics

$$\text{LogLoss} = -\overline{y \log p + (1-y)\log(1-p)}$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

- Log loss rewards calibrated probabilities (clip p)
- MSE punishes large errors; R^2 vs predicting the mean

Recap

- Accuracy misleads on imbalanced data
- Precision vs recall: pick by cost of errors
- AUC for ranking, log loss for calibration
- MSE / R^2 for regression